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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

MICROELECTRONICS – ECC-305 – PBL COMPONENT – I
REVIEW OF RESEARCH PAPERS ON LOW-POWER TECHNOLOGY
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STUDENT DETAILS

Name of Student: _____ **University Roll No.:** _____

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EVALUATOR'S REMARKS

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TITLE OF RESEARCH PAPER REVIEWED

Break-before-Make CMOS Inverter for Power-Efficient Delay Implementation

DETAILS OF RESEARCH PAPER

Title: Break-before-Make CMOS Inverter for Power-Efficient Delay Implementation

Author(s): Janez Puhan, Dušan Raič, Tadej Tuma, and Árpád Bűrmen (Faculty of Electrical Engineering, University of Ljubljana, Slovenia)

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DOI / Paper Identification: Article ID 349131, 11 pages; DOI 10.1155/2014/349131; PMC4265520. Topic assigned: Low-Power Technology, Break-Before-Make (BBM) CMOS Inverter.

REVIEW SUBMISSION

The review includes the following parts.

Background and motivation of the research

Digital systems often need a small, low-precision delay (for example, to align or strobe signals). The usual solution is a chain of inverters. The paper observes that an inverter chain is power-efficient only for small delays: every inverter in the chain drains additional parasitic energy approximately equal to the dynamic energy needed to change its input, so long delays become power-hungry. Direct-path (short-circuit) current is a well-known source of internal dynamic power; in well-designed circuits it is estimated at under 20 % of dynamic dissipation (paper ref. [1], Veendrick 1984), but it can become prohibitive with significant capacitive loads. The motivation is therefore to obtain a required delay with the minimum supply charge (energy) by suppressing direct-path current.

Problem addressed by the researchers

How can a required delay T_p be implemented at the lowest possible charge Q at gate level, where loads are small? Earlier remedies drive the NMOS and PMOS gates of an inverter with separate, time-skewed signals (paper refs. [2]–[5]), but all of them insert additional driver stages ahead of the split inverter; that overhead in area and power is justified only for large capacitive loads, not for gate-level logic. Other gate-level techniques ([6], [7]) address static leakage in nanometre technologies, carry an area overhead, and do not improve internal dynamic power. The paper targets this gap: internal dynamic power of small-load gate-level delay elements.

Low-power technology/technique discussed

Break-before-make (BBM) switching, also called dual-ramp (time-skewed) drive, eliminates direct-path (short-circuit) current. The PMOS and NMOS switching transistors of an inverter are never ON at the same time: during an “isolation time” both are off and the output node is high-impedance, holding its previous logic state on the load capacitance. The technique is dynamic-power oriented (short-circuit component), not a leakage or supply-scaling technique.

Existing approach

A standard two-stage CMOS buffer or serial inverters (paper Fig. 2). The charge per cycle is the sum of a dynamic part in the first stage and a direct-path part Q_s in the second stage, whose input rise/fall times t_r, t_f (Eq. 5) are non-zero:

$$Q_d = (C_{LP} + C_{LN})\, V_{DD}. \tag{4}$$

For a symmetric cell ($V_{TN} = -V_{TP} = V_T$, $\beta_N = \beta_P$) the paper obtains

$$Q = Q_d + Q_s = \left[\frac{2\beta_1 V_{DD}^2}{k_2} + \frac{\beta_2}{6V_{DD}} \left(V_{DD} - 2V_T\right)^3 \right] T_{p,\mathrm{CMOS}}, \tag{9}$$

$$T_{p,\mathrm{CMOS}} = \frac{k_2\,(C_{LP} + C_{LN})}{2\beta_1 V_{DD}}. \tag{10}$$

Both static (direct-path) and dynamic charge grow linearly with the CMOS propagation delay. Here $\beta = (\mu\varepsilon/t_{ox})(w/l)$ and k_2 is a constant; the analysis uses the simple model of Eq. 3 (cut-off / triode / saturation square law).

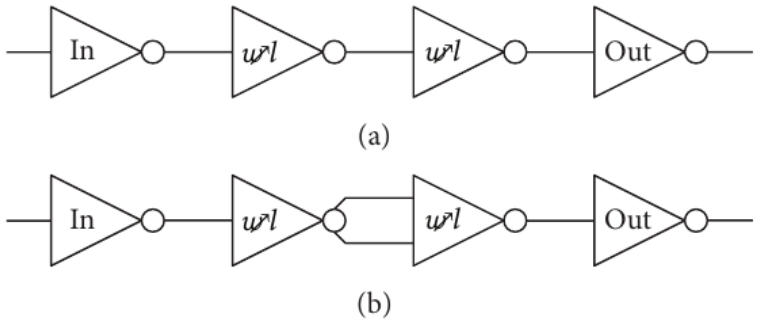


Fig. R1. Standard (a) and proposed (b) delay cell with unit input/output inverters. Reproduced from Puhan et al. (2014), Fig. 9, CC BY licence.

Proposed approach

A bidirectional delay element is connected in series between the PMOS and NMOS drains of the first inverter, producing two time-skewed outputs (p_o, n_o). The next stage is an inverter with split inputs (p_i, n_i) whose outputs are merged (Fig. 1(c) of the paper). Stages cascade without accumulating isolation time, and the overhead is only the delay element, with no extra driver stages.

- R_d **version.** A resistor R_d is the simplest delay element; its RC product with the next-stage gate capacitance sets the isolation time (Eq. 14: $t_{if} = k_8 R_d C_{LP}$, $t_{ir} = k_8 R_d C_{LN}$). Polysilicon R_d is area-hungry and poorly controlled.
- **Transistor versions (paper Fig. 6).** A diode- or triode-connected transistor reduces the output swing by a threshold voltage, needs $V_{DD} > 5V_{Td}$ and gives long information transients, so reduced-swing topologies are judged inappropriate.
- **Full-swing versions (paper Figs. 7, 8).** Level-restoration transistors (controlled by the next-stage output) are added in parallel with the delay transistor. Four two-transistor topologies are compared: PtNt, PfNf, PtNf and PfNt. PtNf (a PMOS delay transistor in triode plus an NMOS level-restoration transistor) is reported as the most efficient.

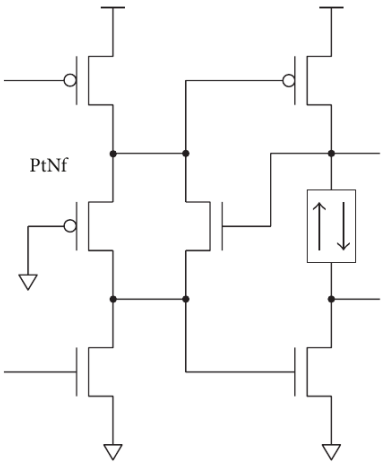
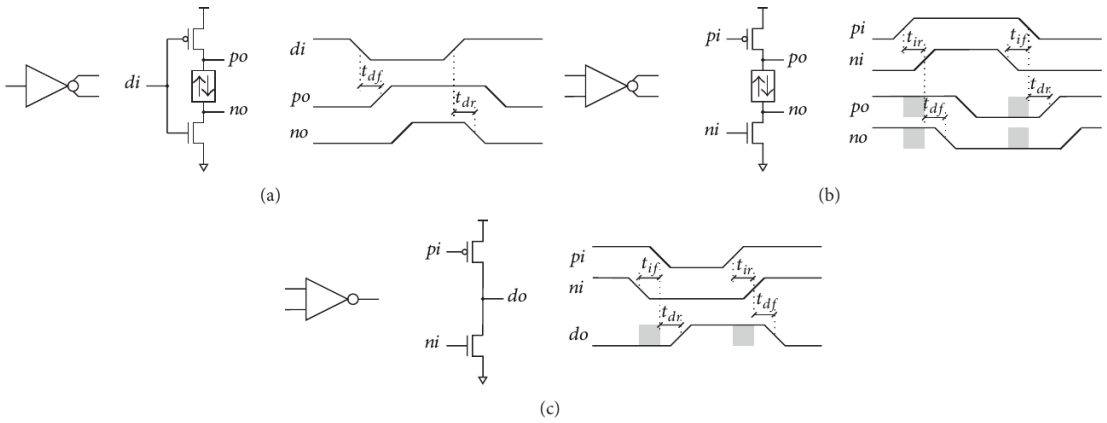


Fig. R2. PtNf full-swing delay implementation. Reproduced from Puhan et al. (2014), Fig. 8, CC BY licence.

Important circuit/device concepts



IGURE 1: BBM inverter structures with time domain responses: (a) CMOS to BBM converter, (b) BBM inverter, and (c) BBM to CMOS

Fig. R3. BBM structures with time-domain responses: (a) CMOS-to-BBM converter, (b) BBM inverter, (c) BBM-to-CMOS converter. Reproduced from Puhan et al. (2014), Fig. 1, CC BY licence.

- **Isolation versus information slopes.** The first edge of a skewed pair (falling edge on the NMOS gate, rising edge on the PMOS gate) is the isolation slope; the second is the information slope. The interval between them is the isolation time t_i, during which the output is high-impedance and no direct path exists. If $t_i \to 0$ the circuit becomes an ordinary CMOS inverter. Isolation times do not accumulate through a cascade.
- **Redefined delays.** t_{dr}, t_{df} are measured from the input information slope to the output isolation slope; t_{ir}, t_{if} lie between isolation and information slopes. The propagation delay (paper Eqs. 1 and 2, symmetric case) is

$$T_p = \frac{t_{ir} + t_{df} + t_{if} + t_{dr}}{2} = t_d + t_i, \qquad T_p = k_1\, T_{p,\mathrm{CMOS}} + t_i, \quad k_1 < 1. \tag{1, 2}$$

- **Charge of the R_d cell.** With $V_{TN} = -V_{TP}$, a symmetric first stage and high R_d, the static charge is $Q_s \approx 0$ and (Eqs. 13–16) $t_{df} = k_6 C_{LN}/(\beta_{N1} V_{DD})$, $t_{dr} = k_7 C_{LP}/(\beta_{P1} V_{DD})$ and

$$Q = \frac{2\beta_1 V_{DD}^2}{k_6 + k_8 R_d \beta_1 V_{DD}}\, T_p. \tag{16}$$

- **Condition for valid BBM operation (Eqs. 17–18).** The isolation slope must end before the information slope begins, i.e.

$$R_d > \max\left(\frac{k_2 C_{LN}}{k_{11} C_{LP} \beta_{N1} V_{DD}},\ \frac{k_3 C_{LP}}{k_{11} C_{LN} \beta_{P1} V_{DD}} \right). \tag{17, 18}$$

- **Figures of merit.** PDP (energy per switching event) and EDP $=$ PDP $\cdot\, T_p$; for a delay line the goal is minimal charge Q at a required T_p, since V_{DD} is fixed.
- **Level restoration.** A reduced-swing node (a V_{Td} drop) is restored by N_f/P_f devices driven from the next-stage output (feedback).

Simulation/experimental methodology

- **Simulator and model.** Synopsys HSPICE with TSMC 0.18 µm / 1.8 V process parameters (proprietary; I did not reproduce it).
- **Circuits compared.** The standard cell of Fig. 9(a) (four inverters) against four BBM delay cells (PtNt, PtNf, PfNt, PfNf) of Fig. 9(b), each with identical standard input and output unit inverters so that fan-in

and fan-out are equal; both unit inverters are counted as part of the cell. An 8-dimensional (standard) or 12-dimensional (BBM) sizing problem was solved for each delay target from 100 ps to 5 ns.

- **Optimiser.** Parallel SADE global optimisation (paper ref. [12]) driven by PyOPUS, on 100 computing nodes (25 Intel Core i5 2.66 GHz processors, 4 nodes each); every run was repeated from several starting points until the global minimum was confirmed.
- **Constraints.** (i) Limits on over/undershoot so the circuit does not rely on Miller (parasitic) capacitance; (ii) steady state reached after $t_{\max}$; (iii) noise-margin constraints from p- and n-substrate disturbances. Without them the optimiser returns "degenerate" circuits with very long transients.
- **Parasitics.** Layout was not in the optimisation loop; instead the layout rules $A_d = A_s = 0.8\,\mu\text{m} \times w$ and $P_d = P_s = 1.5\,\mu\text{m} + w$ were applied for source/drain parasitics.
- **Metric.** Charge consumption (fAs) for one rising and one falling slope; Table 1 below.

Key results reported in the paper

Table 1 of the paper gives the charge consumption of the standard (std.) and BBM delay cells of Fig. 9, measured for one rising and one falling slope, in fAs (femto-ampere-seconds); % is the percentage of the standard cell's consumption. Values are transcribed from the paper; "*" marks a delay target that was not achieved. Conditions: HSPICE, TSMC 0.18 µm / 1.8 V, each cell individually sized for the target T_p.

Paper Table 1. Charge consumption (fAs) of the standard and BBM delay cells.

T_p [ns]	std.	PtNt	%	PtNf	%	PfNt	%	PfNf	%
0.1	34*	45*	—	78*	—	44*	—	97*	—
0.2	26	30	115	87*	—	30	115	62*	—
0.35	32	34	106	34	106	35	109	42	131
0.5	39	38	97	35	90	39	100	38	97
0.75	65	45	69	42	65	46	71	46	71
1.0	77	58	75	49	64	58	75	55	71
2.0	113	86	76	67	59	86	76	93	82
3.5	153	118	77	89	58	118	77	130	85
5.0	186	142	76	108	58	143	77	177	95

- Charge grows approximately linearly with delay above about 1 ns (as Eqs. 9 and 16 predict). PtNf is the most efficient topology; the charge saving against the standard cell is slightly above 40 % (58 % of std. at 3.5 ns and 5 ns). The abstract states that the energy (charge) per delay is reduced "up to 40 %".
- With the exception of very short delays the BBM structure improves the power budget: at 0.1–0.35 ns the BBM cells are no better (PtNf 106 % at 0.35 ns) or do not reach the target delay.
- Third-stage drain currents for $T_p = 750\,\text{ps}$ (paper Fig. 11): the large direct-path current of the standard cell is almost completely eliminated in PtNf. The sizing in the Fig. 11 caption is: *std.* $w_{P1} = 220\,\text{nm}$, $l_{P1} = 370\,\text{nm}$, $w_{N1} = 220\,\text{nm}$, $l_{N1} = 880\,\text{nm}$, $w_{P2} = 880\,\text{nm}$, $l_{P2} = 730\,\text{nm}$, $w_{N2} = 1.05\,\mu\text{m}$, $l_{N2} = 1.32\,\mu\text{m}$; *PtNf:* $w_{P1} = 220\,\text{nm}$ / $l_{P1} = 740\,\text{nm}$, $w_{N1} = 390\,\text{nm}$ / $l_{N1} = 180\,\text{nm}$, $w_{Pt} = 220\,\text{nm}$ / $l_{Pt} = 350\,\text{nm}$, $w_{Nf} = 220\,\text{nm}$ / $l_{Nf} = 950\,\text{nm}$, $w_{P2} = 220\,\text{nm}$ / $l_{P2} = 180\,\text{nm}$, $w_{N2} = 220\,\text{nm}$ / $l_{N2} = 220\,\text{nm}$. The bulk of the delay comes from the third-stage gate capacitances.

Application results. The paper presents a delay line replacing inverter chains (Fig. 12) and a leading-edge pulse generator (Fig. 13). Paper Table 2 gives the charge (fAs) per input impulse, standard versus BBM-based; "*" marks a pulse-width target that was not achieved.

Paper Table 2. Charge (fAs) per input impulse, standard versus BBM-based pulse generator.

No. of inv.	Width [ps]	Std total	Std delay	BBM total	%	BBM delay	%
3	200	171	56	143*	—	17*	—
5	330	210	93	142	68	16	17
7	450	247	130	147	60	16	12
9	570	284	167	153	54	16	10
11	690	321	204	161	50	17	8

The paper concludes that the BBM-based leading-edge pulse generator saves up to 50 % of the total switching energy. The delay-line charge of the BBM version is almost constant (16–17 fAs) because the number of transistors is constant, whereas the standard delay line grows linearly with the number of inverters. Layout parasitics of the cell nodes were not included.

Advantages of the proposed approach

- Direct-path current is practically eliminated in the driven stage, since the PMOS and NMOS are never ON simultaneously (paper Fig. 11).
- Very low component overhead: no additional driver stages; a two-transistor delay element suffices (a six-transistor non-inverting delay cell), so it is usable with the small loads of gate-level design.
- Charge saving of slightly above 40 % at longer delays (PtNf) and up to 50 % of total energy in the pulse generator.
- Delay and short-circuit elimination are obtained in one stage; two output signals give compact functional solutions (for example, an edge detector).
- Isolation times do not accumulate along a cascade, so BBM stages can be chained.

Limitations of the proposed approach

- Not beneficial for very short delays (≤ 0.35 ns: PtNf 106 % at 0.35 ns; target not reached at 0.1–0.2 ns). The authors say so themselves: for logic gates the extra delay and double wiring hardly justify the energy saving. It fits circuits that use the delay (edge-triggered storage, clock distribution).
- Requires dual (two-wire) signals between stages, which costs routing area.
- Reduced-swing transistor topologies need $V_{DD} > 5V_{Td}$ and have long information transients; full swing requires extra level-restoration transistors with feedback from the next stage.
- Optimiser results depend on parasitics unless extra constraints are imposed; the authors could not include layout extraction in the optimisation loop, so layout-induced node capacitances are not accounted for.
- Only simulation results are presented (HSPICE, one proprietary 0.18 µm process); no fabricated-silicon measurements, PVT/Monte-Carlo corners or leakage analysis are reported. The delay is "low-precision" and sensitive to the poorly controlled resistance of the delay element.

Scope for further improvement

- Include layout extraction and parasitics in the optimisation loop (named by the authors as limited only by computing power).
- Verify with process/voltage/temperature corners and Monte-Carlo mismatch, and with modern FinFET/sub-100 nm nodes where leakage dominates.
- Obtain better delay precision using the biased-NMOS (paper Fig. 14) or dynamically biased (paper Fig. 15) cells, whose DC bias power is reduced by dynamic biasing.

- Extend the concept to logic gates (NAND/NOR) in a way that reduces the double-wiring overhead, and to clock distribution and edge-triggered storage elements.
- My own idea: pair BBM with a lower V_{DD} or power gating, since BBM does nothing about leakage.

Relevance to Microelectronics

The paper touches most of what the course covers: (i) the three components of CMOS power, dynamic ($CV_{DD}^2 f$), short-circuit and leakage, and techniques to reduce the short-circuit term; (ii) square-law MOSFET operation regions (Eq. 3) and β, V_T, body effect and channel-length effects; (iii) propagation delay, rise/fall times, PDP and EDP; (iv) transistor sizing trade-offs (long-channel devices create delay cheaply but cost area); (v) level restoration and voltage swing; (vi) simulation-driven design with SPICE and optimisation; and (vii) practical circuits such as delay lines, pulse generators and edge detectors used in dynamic logic and flip-flops.

Technical observations and conclusions

Part A: my own simulation (kept separate from the paper's results). The paper's TSMC 0.18 µm HSPICE model is proprietary, so I rebuilt the Fig. 9 cells in ngspice-42 with a Level-1 model. Its main parameters come from the Fig. 4 caption of the paper: $V_{TN} = 0.5\,\text{V}$, $V_{TP} = -0.5\,\text{V}$, $K_{PN} = 350\,\mu\text{A/V}^2$, $K_{PP} = 90\,\mu\text{A/V}^2$ and $C_{ox} = 8.6\,\text{fF}/\mu\text{m}^2$. Everything else is my own assumption: λ, γ, ϕ, the overlap and junction capacitances, the unit inverters ($W_p/W_n = 860\,\text{nm}/220\,\text{nm}$, $L = 180\,\text{nm}$) and a 2 fF load. Supply 1.8 V, 27 °C, 50 MHz input with 50 ps edges. Q is the charge the whole cell draws from the supply over one input cycle, averaged over four steady-state cycles. Netlists and details are in the ECC-353 report.

Own ngspice results (substitute Level-1 model).

Case	T_p (ps)	Q (fC/cycle)	BBM charge saving
Paper's 750 ps sizing, both cells unchanged, std.	507	67.3	—
Paper's 750 ps sizing, both cells unchanged, BBM PtNf	563	40.3	40.2 % (delays *not* matched)
Re-optimised for $T_p \approx 0.5$ ns, std. / BBM	495 / 505	44.1 / 38.3	13.1 %
Re-optimised for $T_p \approx 1.0$ ns, std. / BBM	1016 / 1006	64.8 / 49.8	23.2 %

Part B: how this compares with the paper. Table 1 of the paper puts PtNf at 90 % of the standard cell's charge at 0.5 ns (a 10 % saving) and 64 % at 1.0 ns (a 36 % saving). With matched delays my model gives 13.1 % and 23.2 %. The shape is the same: the benefit grows with delay and is small near 0.5 ns. The numbers are not. My saving at 1 ns is clearly below 36 %, and the absolute charges cannot be compared because the model, the unit inverters and the load all differ. My optimiser is also tiny next to the paper's 100-node SADE run, so neither of my sizings is likely to be a true optimum.

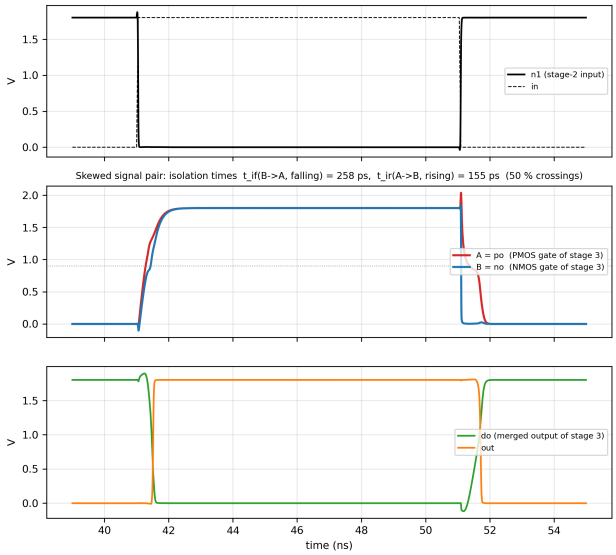


Fig. R4. My ngspice result: skewed pair A (PMOS gate) / B (NMOS gate), isolation edge first, information edge second (paper's 750 ps PtNf sizing, substitute model).

Part C: what I noticed. The skew is easy to see in the waveforms. With the paper's 750 ps sizing the isolation edge (A rising or B falling) arrives 155–258 ps before the information edge, and the two conducting windows never overlap. What surprised me was the crowbar charge. In the conventional cell's third stage it is only about 0.7 fC per cycle (a rough $\min(i_P, i_N)$ estimate) out of the 15.4 fC that stage draws, so I cannot say the saving comes mainly from removing short-circuit current. A good part probably comes from BBM letting P_2 and N_2 stay small. Node A also overshoots to 2.04 V through Miller coupling, which is exactly what the paper's constraints are meant to suppress. Finally, taking the delay element out is not a fair ablation: T_p drops from 563 ps to 155 ps, which only shows that the P_t resistance is what makes the delay.

Part D: conclusions. BBM switching does lower the charge needed for a given delay. The paper reports slightly more than 40 % at 3.5 ns and above, and nothing gained below about 0.35 ns. My substitute-model runs show the same trend, with 23 % saving at about 1 ns and 13 % at about 0.5 ns. For ordinary logic gates it is a poor deal, since the delay is not wanted and the second signal wire costs area. My results depend on the Level-1 model (no velocity saturation, no leakage), the small load and a short optimisation, so I treat them as a trend check, not a reproduction.

References

[1] J. Puhan, D. Raič, T. Tuma, and Á. Bűrmen, "Break-before-Make CMOS Inverter for Power-Efficient Delay Implementation," *The Scientific World Journal*, vol. 2014, Article ID 349131, 11 pp., 2014. doi:10.1155/2014/349131. https://pmc.ncbi.nlm.nih.gov/articles/PMC4265520/ (primary source; CC BY).

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[3] N. H. E. Weste and D. M. Harris, *Principles of CMOS VLSI Design: A Circuits and Systems Perspective*, 4th ed., Addison-Wesley, 2011 (cited in [1] for the square-law model; not independently consulted).

[4] ngspice, open-source mixed-level/mixed-signal circuit simulator, version 42, https://ngspice.sourceforge.io/ (used for my simulations).